

Cisco Umbrella

Lec-1

Complete Training on

- Umbrella DNS platform
- SASE
- Umbrella SIG

monthly

Sec1 - Umbrella Architecture and Deployment

Sec2 - Identities and policies

Sec3 - virtual Appliance and AD Integration, Meraki, ASA, SIEM

Sec4 - Intelligent proxy and file inspection

Sec5 - Roaming computers and mobile users

Sec6 - Umbrella SIG - CDW, SaaS, DLP, IPS

Pre-Requisites

• Basic Networking Knowledge

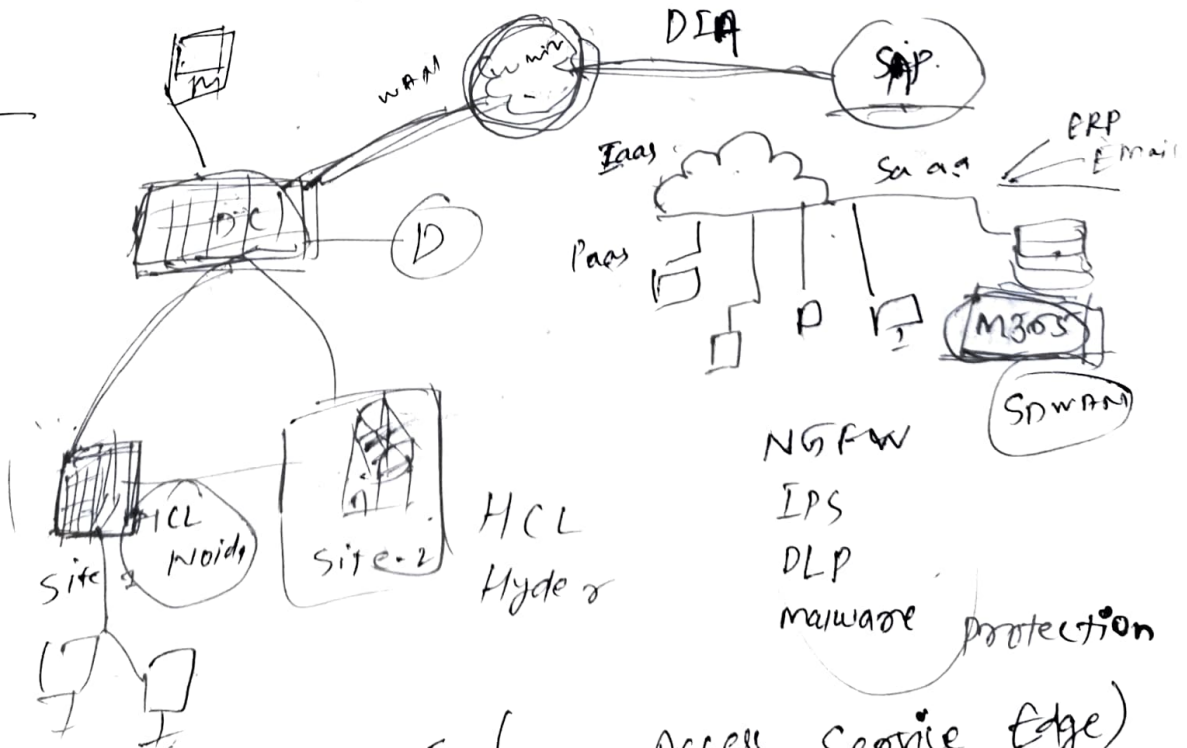
- IP Addressing
- DNS protocol

• Basic Security Knowledge

- Access control - policy, granular details
- Firewalls
- web proxy
- Policy management



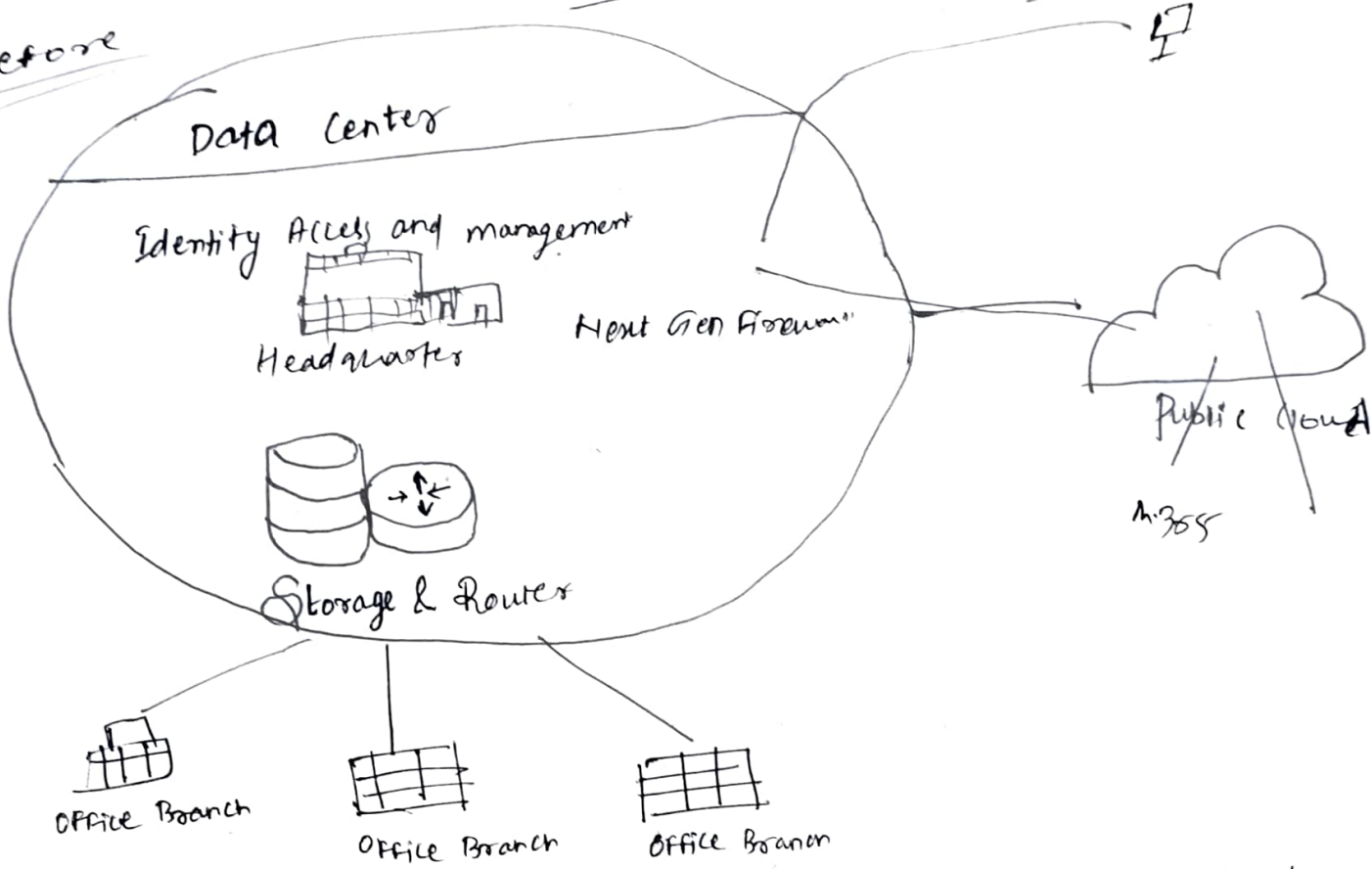
Before



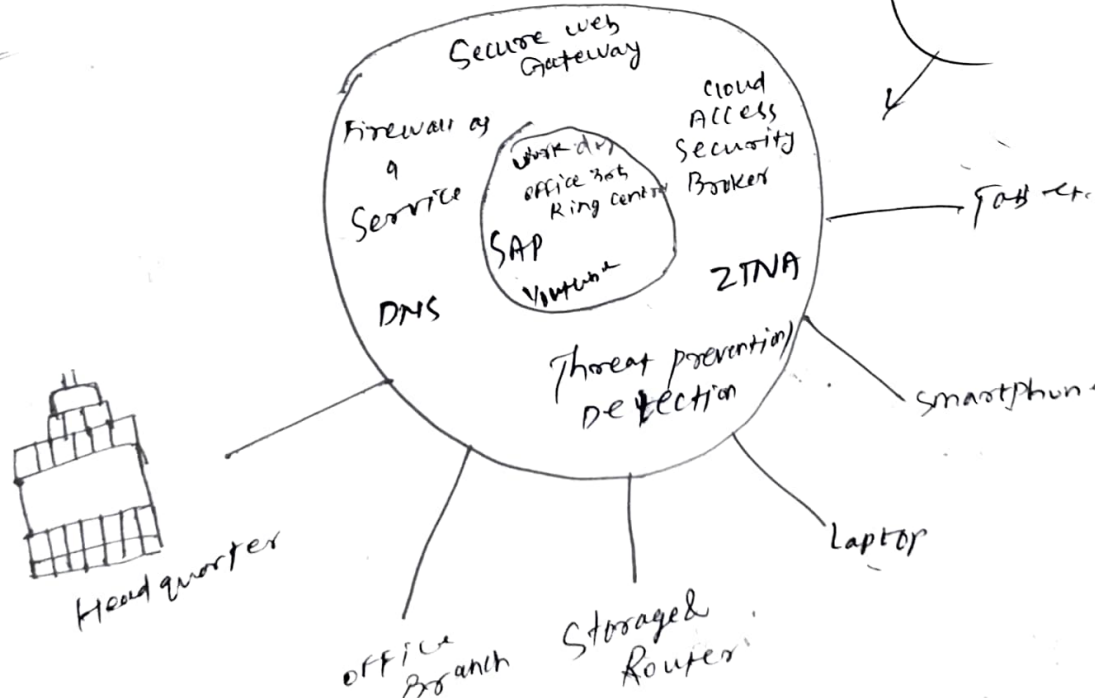
SASE (Secure Access Service Edge)

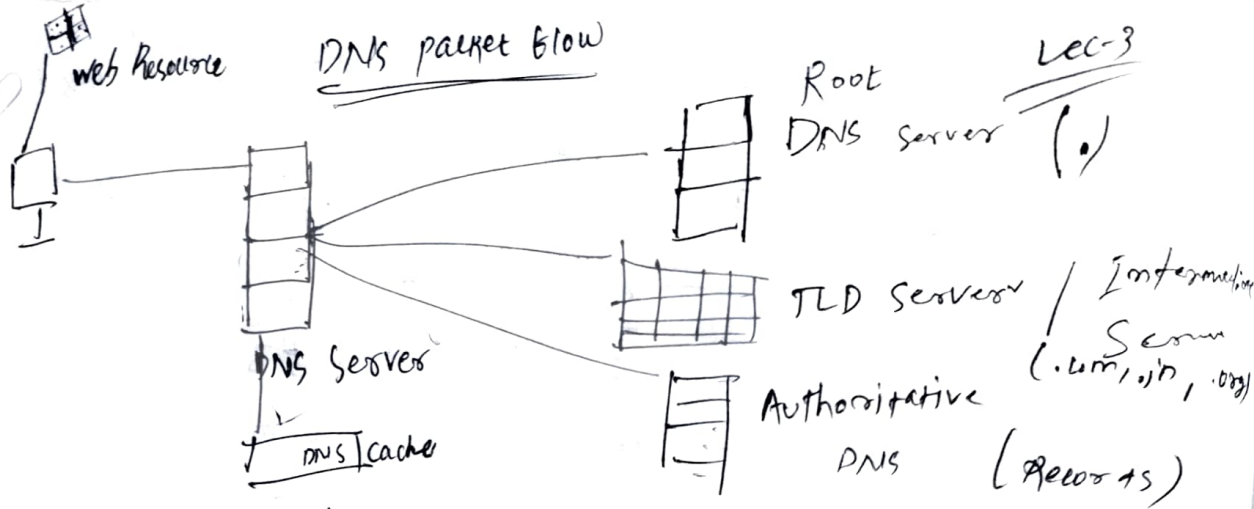
GARTEN ER

Before



Now





Domain / URL $\rightarrow$ IP Resolution

A Record

IP $\rightarrow$ Domain / Hostname

PTR Record

Email Server Information

MX Record

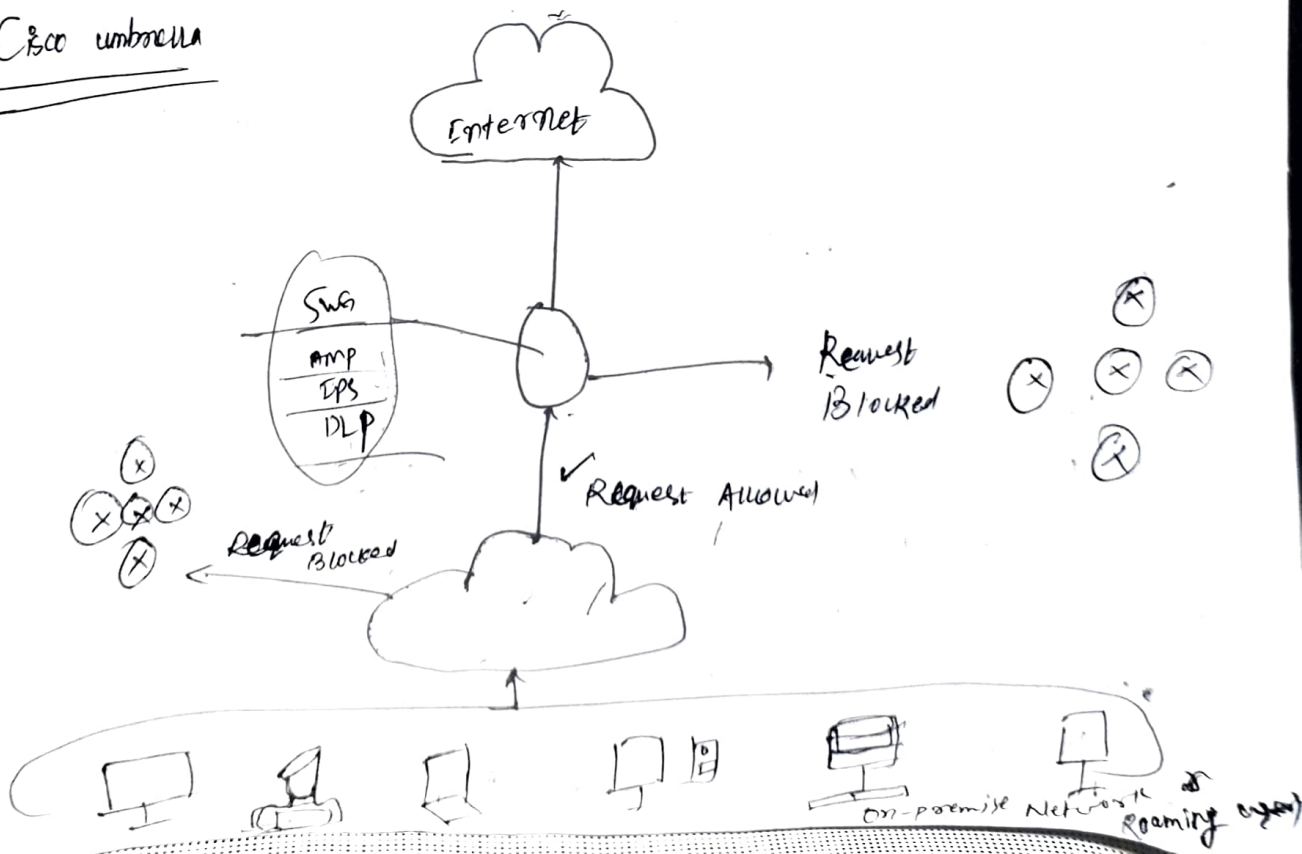
Canonical Name

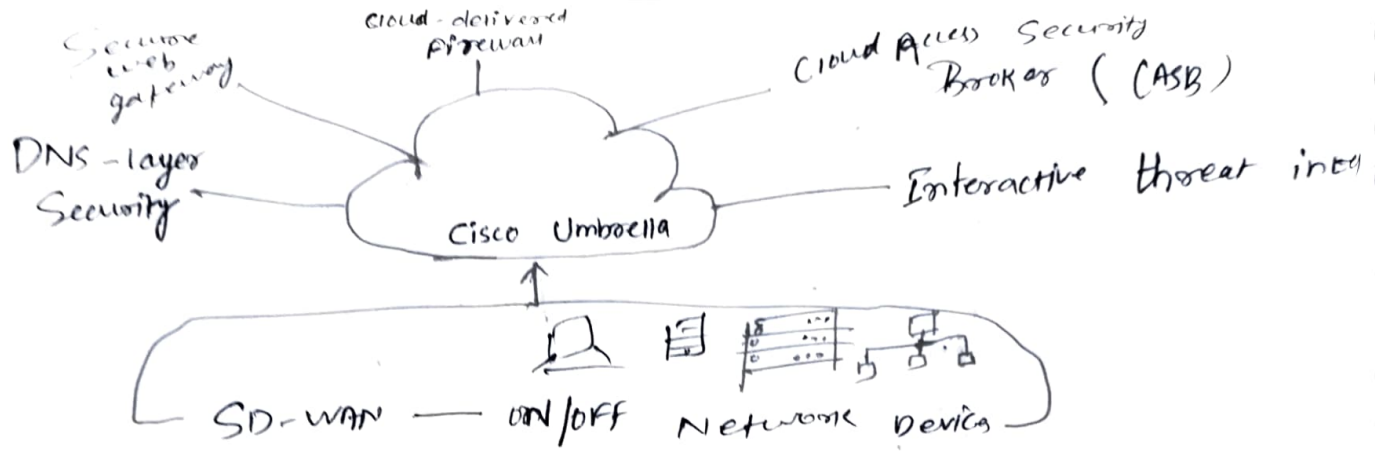
CName Record

IF someone spoofing
then there will be a problem

Lec-4

Cisco Umbrella





i) DNS layer security: DNS Resolution with security

ii) Secure web gateway: → connection flow security

- filters the content in order to secure network from malicious content, malicious files being downloaded
- SSL decryption
-

iii)

Based Gateway protocol

1. DNS Layer Security:

- First line of defense
- Blocks malicious domains before a connection is made
- protects users on-network and off-network
- Stops phishing, malware, and C&C callbacks at the DNS level

2. Secure web Gateway (SWG)

- Inspects web traffic
- Enforces acceptable use policies
- Blocks malicious URLs, risky categories, and unwanted content
- Works with proxy and PAC files.

3. Cloud-Delivered Firewall (CDFW)

- Layer 3 & 4 Firewall in the cloud
- Controls traffic using IP, port, and Protocol
- Useful for restricting outbound traffic without on-prem firewalls

4. Cloud Access Security Broker (CASB)

- provides visibility and control over cloud apps (SaaS)
- Detects risky apps and shadow IT
- Enforces policies on sanctioned apps like Microsoft 365, Google Workspace

5. Interactive Threat Intelligence

- Real-time visibility into threats
- powered by Cisco Talos
- Shows domain reputation, IP risk, malware details, and attack timelines
- Helps SOC teams with investigation and incident response

Lec 6 - Umbrella packages:

There are four main Cisco Umbrella security packages available, designed to fit different team needs: DNS Security Essentials, DNS Security Advantage, SEG Essentials, and SEG Advantage.

	DNS Security Essentials	DNS Security Advantage	SEG Essentials	SEG Advantage
① <u>Secure web Gateways</u>				
➤ Proxy and inspect web traffic (incl. decryption of SSL (HTTPS) traffic)	-	Partial	Full	Full
(i) Enable web filtering by domain or category (SaaS filtering by URL)	Full	Full	Full	Full
(ii) Create custom block/allow lists of domains	-	-	Partial	Full
(iii) Create custom block/allow lists of URLs	-	-	Full	Full
(iv) Block files based on AV Engine and malware defense	-	Partial	Full	Full
(v) Use malware analytics (Sandbox) on suspicious files	-	-	500/dy	Unlimited
② <u>Cloud access security broker</u>	Partial	Partial	Full	Full
(i) cloud app discovery, risk scoring, blocking or activity controls	-	-	two arm	Full
(ii) Scan and remove malware from cloud-based file storage apps	-	-	-	-

Information Security

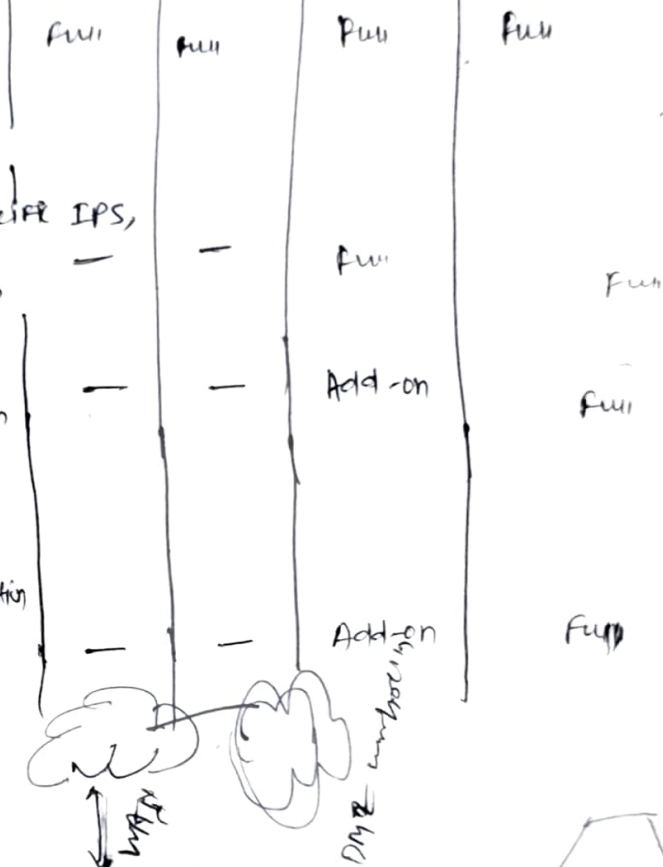
Excludes domains with malware, phishing, botnet, or other high risk items

cloud delivered pipeline

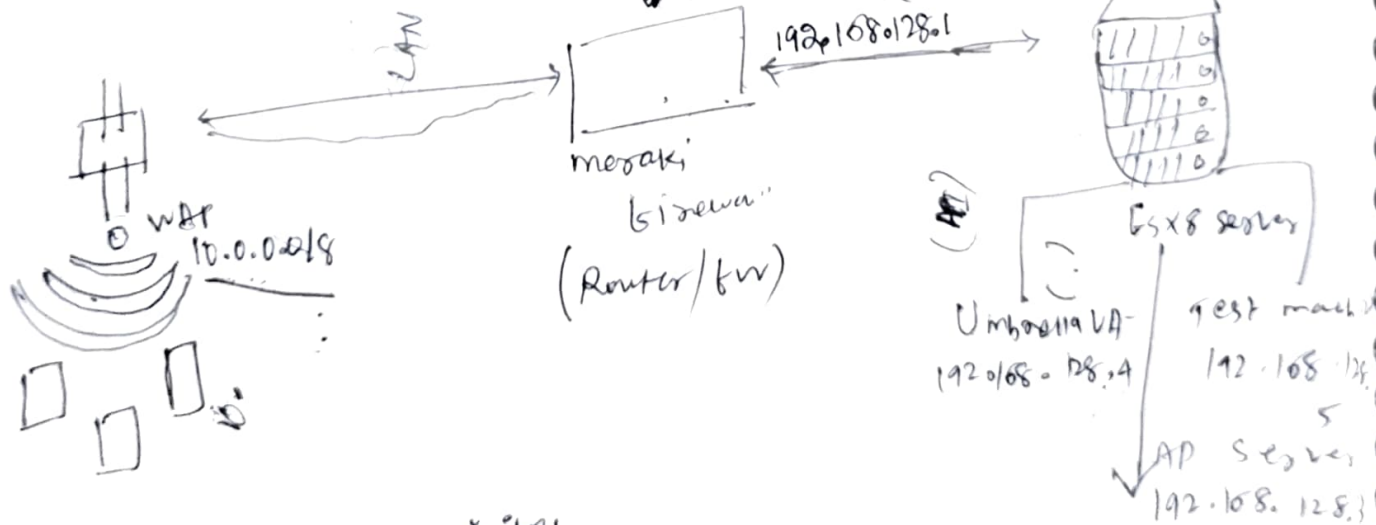
- 1) Create L3/L4 policies to block specific IPs, ports, and protocols
- 2) Leverage L7 protection including an intrusion prevention system

⑤ Multimode cloud data loss prevention

Enable inline and out-of-band inspection and blocking capabilities to protect sensitive data



lec-7 - Lab setup



Lec-8 - Deployment Pre-Requisites

Acceptable Use policy - It defines what is acceptable or what is permitted and what is denied

Q. Is social media site is permitted when you are on the organization's network, wherein can you access facebook, twitter, instagram, snapchat, can you use sites like firestorm and

④ ammunition?

ONS Layer Security

Block domains with malware, phishing, botnet, or other high risk items

④ cloud delivered firewall

i) create L3/L4 policies to block specific IPS, ports, and protocols

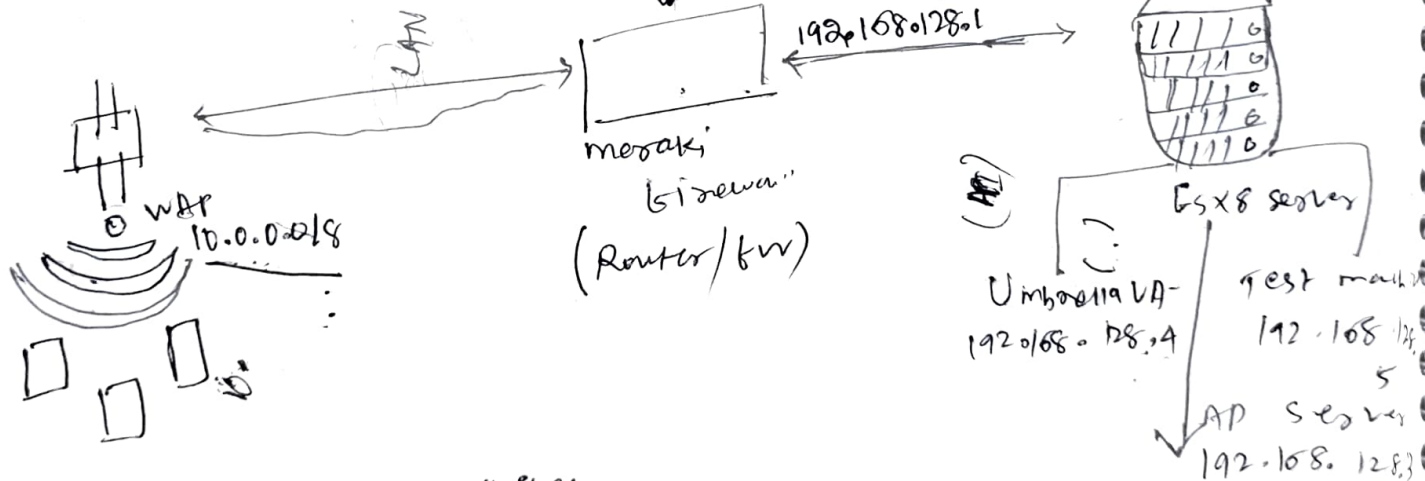
ii) Leverage L7 protection including an Intrusion prevention System

⑤ Multimode cloud data loss prevention

Enable inline and out-of-band inspection and blocking capabilities to protect sensitive data



Lec-7 - Lab Setup



Lec-8 - Deployment Pre-Requisites

Acceptable use policy - It defines what is acceptable or what is permitted and what is denied

Q. Is social media site is permitted when you are on the organization's network, wherein can you access facebook, twitter, instagram, snapchat, can you use sites like flipkart and

⑥ ammunitions?

Can you use recreational sites like games and spend some of your free time out there?

Deployment Workflow:

Add Networks and identities } Point DNS To Cisco Umbrella } Setup policies } Mature & Improve

Section 3 - Initial Setup

Lec-9 - Identities & Policies - Identifying the Source

Identity provisioning

- Networks - Add IP Networks and Internal Subnets
- Users and Groups - Requires AD Integration, on-prem or Azure AD, including SAML
- Network Tunnels - IPsec based site-to-site tunnels
- Sites - External sites from organization
- Roaming computers - Agents on Laptop (AnyConnect module or Roaming agent)
- Mobile client - iOS and Android based devices
- Network Devices - Directory integrated devices

1) Core Identities: ~~not~~ above ones

2) Configuration (Domain management, Sites and Active Directory, Internal Networks, Root Certificate, SAML configuration, Service Account exceptions)

3)

3) Policies:

Management: (DNS, Firewall, Web, DLP policies)

4) Reporting: Policy Components: IPS signature lists, Destination lists, Content categories, Application setting, Tenant controls, Schedule setting, Security setting, Block page appearance, Integration setting, Data classification, etc.

Core Reports: Security Activity, Activity Search, App discovery, Top Threats

Additional Reports: Total Requests, Activity Volume, Top Destination, Top Categories, Top identities, cloud malware, Data Loss Prevention

Management: Exported Reports, Scheduled Reports, Admin Audit Log

Investigate: on security events

Smart Search, Pattern Search, API keys

Admin panels:

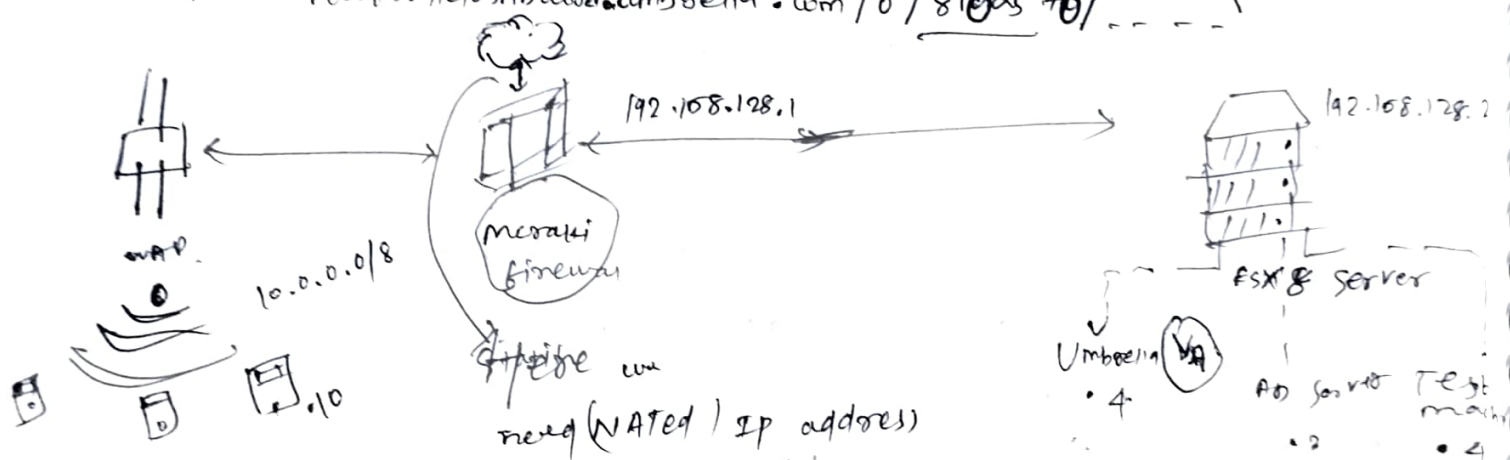
Account, User Roles, Log management, Authentication, Bypass user

Bypass Codes, API keys, Licensing

OID (Organizational Identifier)

In URL: <https://dashboard.umbrella.com/o/8102570/>

Access
to block
sites
(bypass umbrella)



→ If it is static then we can assign

→ If dynamic, we need to download umbrella client

Enter Umbrella console: Copy public IP address (103.182.221.64)

click on add

Add a network

IPv4: 208.67.220.220 and
" 222.222

IPv6: 2001:19:35:35 and
" 53:53

Network name: Security-SouthAfrica-Holds

OS only OSes only Mixed IPv4 & IPv6

IPv4 Address: 103.182.221.64 / 32 Mark value

Cisco Meraki (Security & SD-WAN - Meraki Dashboard)

1.23

Security & SD-WAN

Configuration

WAN < primary DNS (208.67.222.222)
Secondary DNS (208.67.220.220)
[update]

DHCP

customer nameservers

208.67.222.222

208.67.220.220

(Save changes)

Lec-11

Dynamic DNS Setup

Add - Learnmore - 5. Dynamic IP updates:

→ windows IP updater

→ mac IP updater

→ Linux IP updater

In Cisco Umbrella, a Dynamic DNS (DDNS) user is used to identify and track a network whose public IP address changes frequently.

Instead of relying on a fixed public IP, Umbrella uses a DDNS user + token to keep the location updated.

Why Umbrella needs it:

Umbrella enforces security policies based on locations (IP-based)

If ~~the~~ ISP gives a dynamic public IP, Umbrella would lose visibility when the IP changes.

So:

→ DDNS user automatically updates the current public IP

→ Umbrella always knows "this traffic belongs to the location"

How it works:

i. You create a Dynamic DNS user in Umbrella

ii. Umbrella generates a DDNS update URL / Token

iii. You configure it on:

• Router • Firewall • Server • Umbrella VA (Virtual Appliance)

iv. When public IP changes:

• Device calls the Umbrella DDNS update URL

• Umbrella updates the location's IP automatically

Configure it in Umbrella:

• Admin → Locations → Add Location

• Choose Dynamic IP

• Umbrella creates a Dynamic DNS user

OpenDNS updater

OpenDNS username or email: - - -

password: - - -

Q. What do we need to do if we want to see internal IP in the Cisco cloud umbrella?

A: The answer to this issue is to install something called as a virtual appliance or VA inside your network. So that all of these devices can send their DNS request to that local resolver.

In that way, we would be able to identify accurately and pinpoint who the source or the internal IP address.

Go to ESX Server

virtual machine

Secureview - AP

create / register vm

i) Select creation type -

Create a new vm

Deploy a virtual machine
from an OVF or OVA file

Register an existing vm

ii) Select OVF and VMDK files

Enter a name for vm

Secureview - VA

iii) Select storage

Standard persistent storage

Select a datastore for the virtual machine configuration files and all of its virtual disks

Name	Capacity	Free	Type	This Partition	Access
VM storage					

(iv) Deployment Options

Network mapping
Disk provisioning
Power on automatically

Network VM Network v
• Thin o Thick
☒

(v) Ready to complete

Product - VMware virtual Appliance (premise)
vm Name -
Files -
Datastore - VMStorage
Provisioning type - Thin
Network mappings - Network VM Network
Guest OS Name - Linux 5.4.6-124-generic

2

virtual machine → Securewise VA
Enter username and password

Address
Port
Pass

IP address & config VA show

(It will show the current version or the configuration that is living on this machine)

\$ config va name Securewise-VA1
Config va show

Local VMs
IP address
OSSEC: disabled

Internal domains count: 2

Resolvers: 208.67.222.226 208.67.222.220

SSH access: disabled

HTTPS status: disabled

SSL certificate start time: NA

SSL certificate expiry time: NA

2 users: NA

Common Name: NA

Subject Alternative Name: NA

Per-IP based Rate Limiting: NA

Primary address: 208.67.222.220

IP: 208.67.222.220

IP: 208.67.222.220

Gateway: 192.168.1.1

IP: 208.67.222.220

IP: 208.67.222.220

IP: 208.67.222.220

192.168.128.1 \$ config via interface ens160
port name adapter name

192.168.128.4
ip address

255.255.255.0
subnet mask

192.168.128.1

192.168.128.1
Gateway

(Local DNS queries from this domain are routed to our domain controller.)

192.168.128.1 \$ config localdns add 192.168.128.3
localdns server 192.168.128.3 is added successfully

LEC-14 VA Installation verification

Configuration

→ Sites and Active directory

Securix - VA2

192.168.128.4

This VA isn't redundant with 1st site (2nd VA required per site) for more info. and steps to live VA import imported 17 minutes days version 3.4.3

Configuration

Domain management

Internal domains (3)

Add → Add New Bypass Domain to Server

o internal o External domains & IPs

Domain:

Securix.org

Description:

Applies To:

All Sites All Devices

Configuration:

Internal networks: Add a New Internal Network

Name: Securewise-South-LAN

IPv4 Address: 192.168.128.0 / 24

Internal Network Association

o site o Network o Network Tunnel

Sites

Select a site ✓

Securewise-Southoffice

Name: Securewise-Isolated-WLAN

IPv4 Address: 10.0.0.0 / 8 (16384K)

Internal Network Association:

o site o Network o Network Tunnel

Sites

Select a site ✓

Securewise-Southoffice

Bundle

Next step: The next step, I need to do is bundle all of these up into a site, ~~right?~~

check: Sites & Active Directory :

site: Default site

Securewise-Southoffice

Setting

Add New site

Site Name

NOTE: When only the Domain controller is integrated with Umbrella, but end-user PCs are not, this happens:

User PC → Domain Controller DNS → Internet

what happens:

Umbrella Only Sees:

DC = Umbrella

But not:

User = Website

we need to change the configuration in a firewall, so that all the ^{DNS} traffic is forwarded to umbrella VA.

Meraki: wireless

Client IP and VLAN:

custom DNS servers

content filtering

Enabled

Disabled

192.168.128.4

It 2nd one present

Remember

AD Server also
can act as DHCP Server

Then in DNS Setting add
DNS as a VA IP

Now we can see what are internal IP of the user in the dashboard of Cisco Umbrella.

Sec-5 - Active Directory (AD) Integration:

Active Directory is integrated with Cisco Umbrella to provide user and group-based identity, enabling granular security policies and accurate user-level reporting instead of only IP-based control.

We can create groups like CEO, IT, and HR

i) you need to get something called as a configuration script and this script would need to be executed on your AD servers.

ii) once the script is executed by using admin permission, it directly registers

NOTE: If you have multiple domain controllers in your environment, you need to execute that configuration script on the global catalog server or master domain controller. We don't need to run it on multiple machines ~~and~~ again, the connector service, it needs to be executed on the main domain controller service itself.

However if ~~we~~ have multiple domain like securewise.org, Securewise.com, Securewise.in, and securewise.local.

If we have things like that inside your network, ~~you~~ would need to ^{install} ~~install~~ all those components on each or the global catalog server.

Corp - umbrella console (Securewise and AD)
Deployment → Sites and AD → Download → AD Component

→ windows configuration script for domain controllers
(based on our requirement)

i) After download, Create one ^{DNS} Open connector account
it contains: (1) open DNS - windows ConfigurationScript
(2) Open DNS - window - Service

ii) open AD (Win + Q)

AD users and components

→ Securewise.org → Users → New → (user or group) _{req}

in Downloads folder

cscript OpenDNS - windows ConfigurationScript.wsf

would you like to register this Domain Controller (y or n)? y

iii) execute this particular service } execute the setup (install)

(These above 2 scripts is used to sync user accounts and group changes whenever you have modification things like that)

Lec-16: AD Integration with Local DNS:

ESX host → SecureWire-Va & config local dns add (ip-address) domain-controller

Section-6: Policy Management:

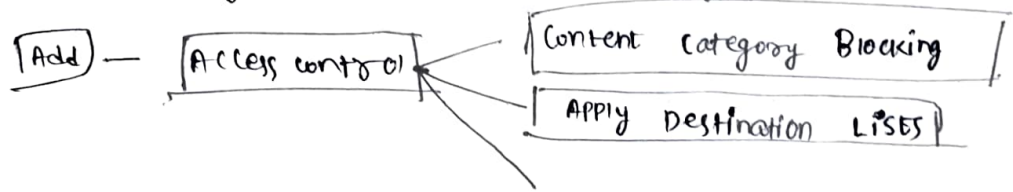
Lec-17: Policy deployment

umbrella policies

- Applying security controls to Identities
- Allow and Blocking actions
- Have multiple policies
- Top Down Approach
- Build policies from Bottom Up

Policies > Management > DNS policies

☒ AD Allow handling sites for IT



Lec-19: Whitelist & Blacklist

~~White~~ Add

- ☐ unchecked Application Control
- ☐ unchecked File Analysis
- ☐ unchecked intelligent proxy
- ☒ Check Allow only mode

(we create rule for admin who has least access)
whitelist mode
→ whitelist sites are allowed

Rule for particular users → AD Users → SecureWire-0614
→ custom → ~~Select all~~ ~~block everything~~ → Add new...

Administ

Destination

google.com

cisco.com

default
block page
Appearance

Admin
Policy

check using users } in the browser

Task: whitelist linkedin for all and block snapchat for all regardless of AD groups and users.

Access Control

Choose: ~~Networks~~

- Access control

Apply DST list

Block Threats

ADD New list → Name
Blocked (action)
Snapchat.com (URL)

Policy Name - SW-BLLIST

(Similarly whitelisting)

Policy Tester:

used to check find the rule that is responsible for blocking or allowing.

Lec: 20 - Global whitelists & blacklists

Policy components → Destination lists

Global Allow List

Global Block List

Lec: 21 - Customizing the block page appearance

Policy components - Block page Appearance

Add - (SW Block page) - Allow blocked users to contact an admin from the block page
show a block page with a custom message
* [domain] is blocked by the organization. please contact the support at [email address]

main page is also possible

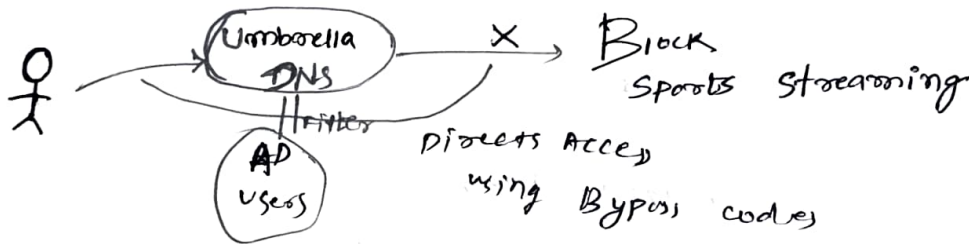
Now, we can go to one of the DNS policy, ^{change} Block page appearance in the policy

Lec-22: Application Management

Policy > management > ~~Application~~ API policies

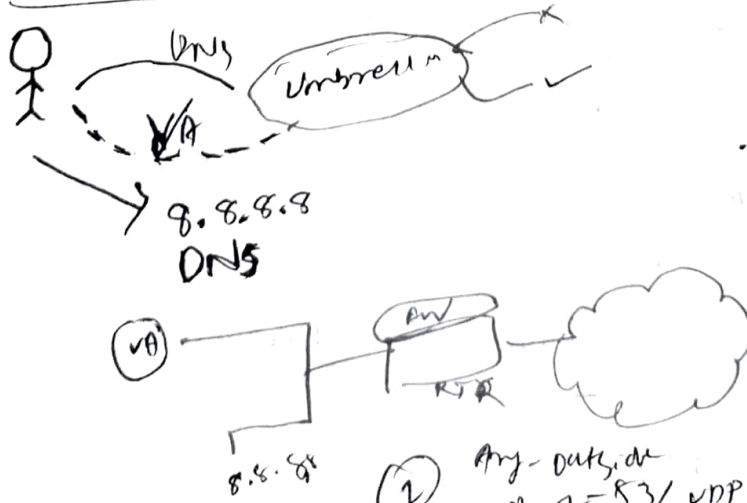
- > Application Setting Applied
- Default setting
- edit / Save
- New setting
- Block applications

Lec-23: Exceptions & Exemptions:



Bypass codes are temporary codes that allow a user to skip umbrella protection for a short period of time when a website is blocked.

Lec-25: prevent umbrella Bypass?



- (1) Any outside port 53/UDP - Umbrella Server

- (2) Permit UDP/53 from VA deny any UDP/53 from any

Policy > management >

Policy component >

Destination Lists

Dest List

CEO-whitelist

Blocked Allowed

Facebook.com

Bypass codes
in All policies

St-2

Policies >

All Policies

custom-blacked pages

Bypass codes

new code

Search Dest List

CEO-whitelist

St-1

create a user

Accounts

Addin

Super
user

Superuser

Black page
matters

Superuser @supanidre.org

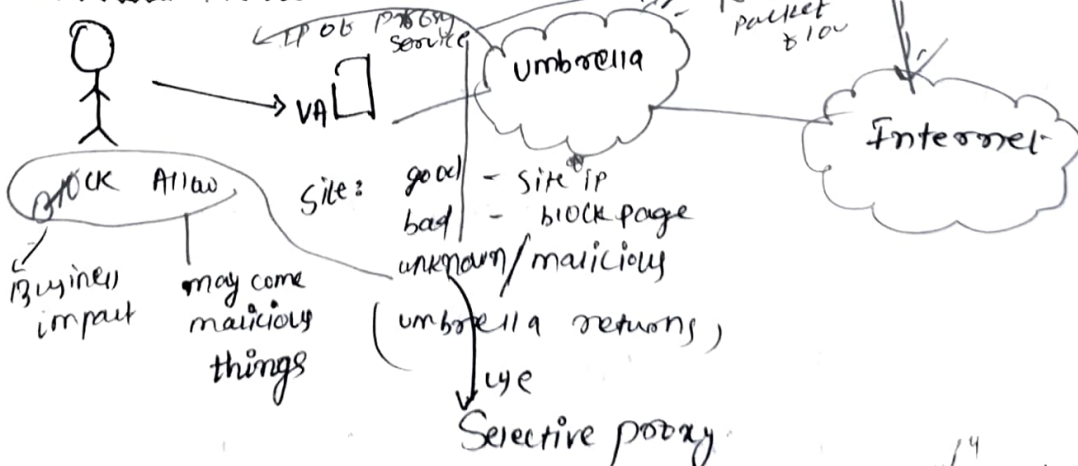
password

password

Email address

Sec-8

Intelligent proxy & File Inspection



if good website == good;

return ip(of site);

else if (website == bad)

return "block page";

else

return "block proxy service";

if something found malicious
it will break the connection

lec 29: Intelligent proxy deployment

Intelligent proxy is a feature in Cisco Umbrella that automatically decides whether web traffic should be inspected or not, instead of proxying all traffic.

Q. Why does Umbrella use intelligent proxy?

A: Proxying every website would:

- Slow down browsing
- Break some applications
- Waste inspection resources

So Umbrella asks:

Is this website risky enough to inspect deeply?

If yes → Send traffic to the proxy

If no → allow it directly

Q. How intelligent proxy works?

i) User requests a website

ii) Umbrella checks:

- Domain Reputation
- Content Category
- Policy Setting

iii) Decision:

Low risk → DNS-only decision (no proxy)

Medium/High risk → Traffic is redirected to the Umbrella proxy

iv) Proxy performs:

URL filtering, Malware inspection, File inspection, SSL decryption

Q. What kind of traffic goes via intelligent proxy?

A: Traffic that needs deeper inspection, such as:

- Unknown domain
- Newly registered domains
- Risky categories (malware, phishing, crypto-mining, etc.)
- File downloads
- HTTPS traffic requiring content inspection

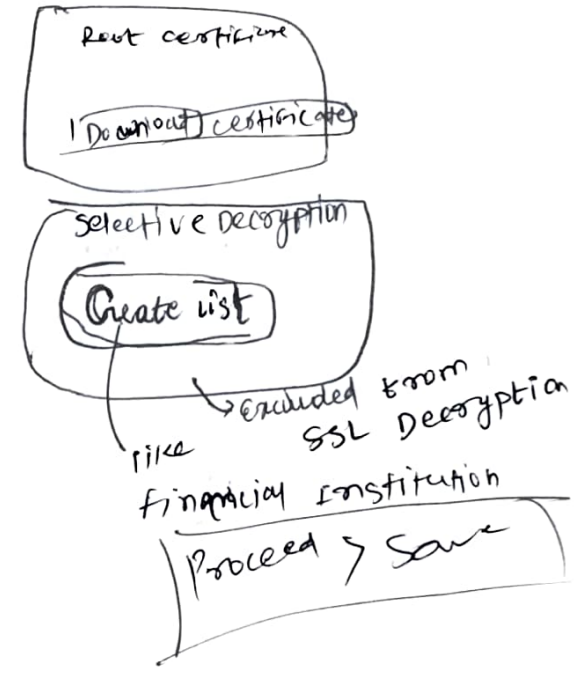
Q. Where is Intelligent proxy used?

- Umbrella Roaming Client
- Anyconnect Secure Client (Umbrella module)
- SD-WAN Secure web gateway

policies > management > DNS policies > Advanced settings
 (Default policies)

☒ Enable Intelligent proxy
☒ ~~Enable~~ SSL Decryption

URL: proxy.Openshifttest.com/indextest.html
 tells whether your umbrella uses
 Intelligent proxy or not



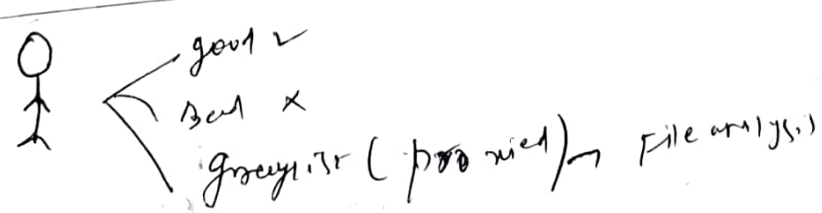
Checking:

Reporting > Core Report > Activity Search

Responses:
☐ Allowed ☐ Blocked ☒ Selective proxy
 proxy logs

Lec. 31:

File Analysis:



Internet ☒ malicious

(AU)

Intelligent proxy (AMP) (while downloading if it finds file is malicious then it will block)

known bad hashes of malware

- Encrypted (x)
- Zipped / Archived (compressed) - inspect upto 16 iterations / layers

Lec. 32: File inspection and implementation

policy > management > DNS policies > Default policy

☒ File Analysis

Keynote 3 SSL Decryption enabled

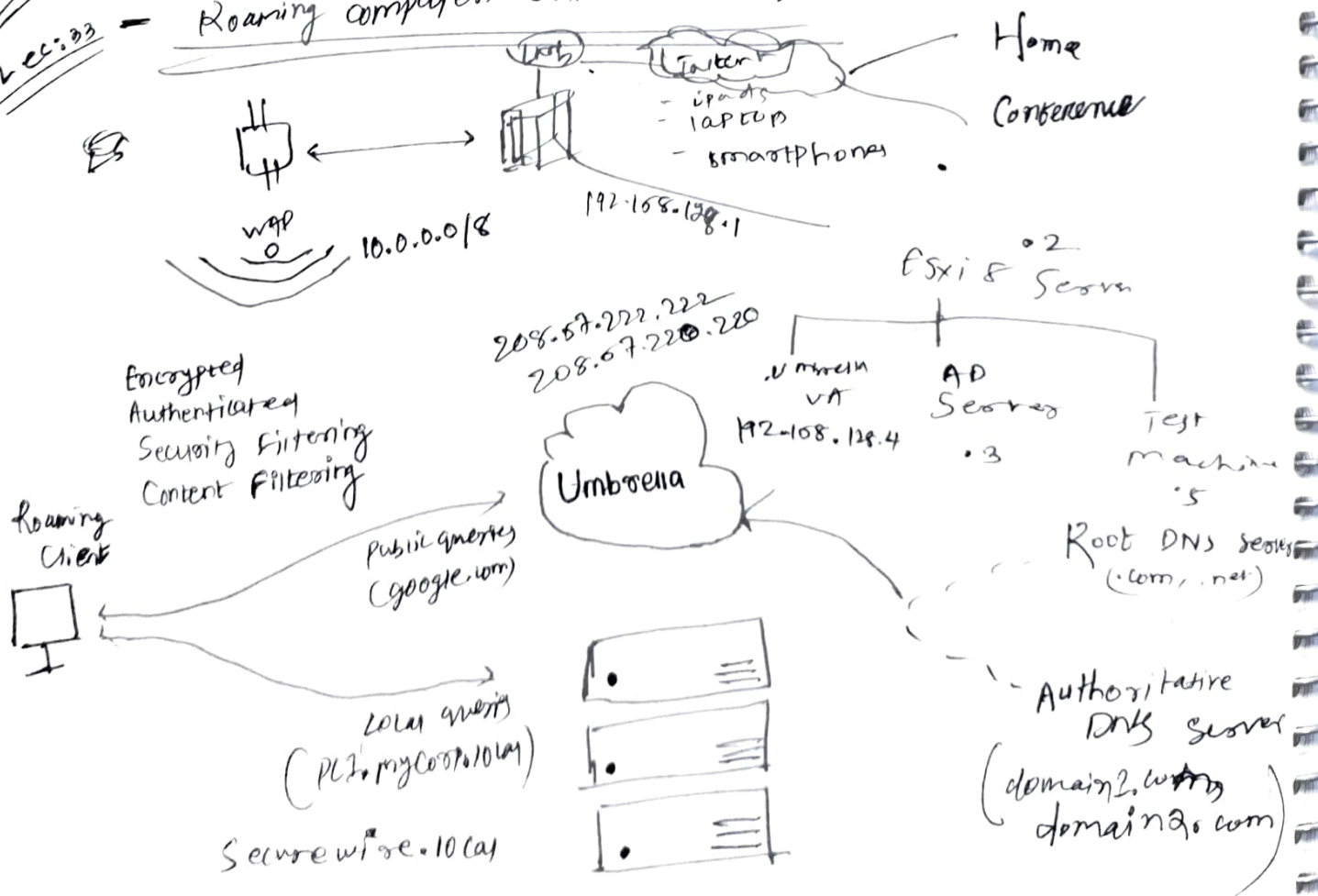
www.telerik.com/downloads / malware samples

If you test - files won't ^{be} downloaded

⇒ Secure proxy is downloaded only participates in greatest site.

Sec 9: Roaming Computers & mobile devices

Lec. 33 - Roaming computers Introduction;



Umbrella For Roaming Users

Roaming client - Only DNS Layer Security

Roaming Security module for Anyconnect

- DNS Layer Security + SWG (SIG)

→ Roaming secure client in Cisco Umbrella is a lightweight agent installed on end-user devices (like laptops / desktops) to protect them when they are outside the company network.

→ It ensures the device is still protected by Umbrella even if it is:

- Connected to home wifi
- Using mobile hotspot
- On public Wifi
- Not connected to corporate VPN

Q. Why Roaming Secure client is Needed?

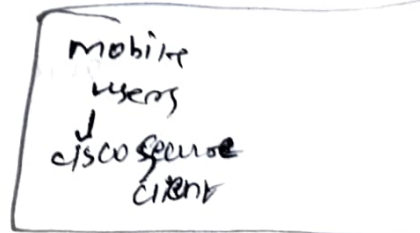
Normally, Umbrella protection works through:

- Internal DNS (inside office)
- Network Forwarding / virtual appliances

But when users leave the office, DNS traffic doesn't pass through corporate network.

So, Roaming Secure client forces DNS traffic to go through Cisco Umbrella cloud.

Q. How it works?



1. User types a website

2. DNS request is intercepted by Roaming Secure client.

3. Request is securely sent to Cisco Umbrella cloud

4. Umbrella checks : Is it malicious? Is it blocked by policy?

5. If same → allowed

If malicious → Blocked

Lec-34s Roaming Agent Deployment

Step 1: Deployments > Roaming computers > Roaming client

Download: windows, MAC

Folder: OpenDNS - URC - win - 3.0.343 JSON file
orginfo.json

```
{  
  "organizationId": "8102570",  
  "fingerprint": "8a08431d7d3276857ea06365457d93d7",  
  "userId": "12136630"  
}
```

Setup insra

After installing ~~this~~ set up file
Device will ~~appear~~ in this ~~location~~ path.

Step-2: DNS policies (It follows the default path)

- ☒ Access control
- ☒ Block threats
- ☒ Content category Blocking
- ☒ Security category Blocking
- ☒ Apply destination lists
- ☒ File Analysis
- ☒ Application control

^ Adv setting ✓

- ☒ Enable intelligent proxy
- ☒ SSL Decryption

next

Step-3 protect: > Roaming Computers

Select Setting > Securewise Setting

Step-4 create new Setting

Lec: 35 - Roaming Agent Module for Anyconnect Deployment

Secure Client

File - Cisco-secure-client-win-5.0.01242-predeploy-k9 > Setup

- ☒ core & Anyconnect VPN
- ☒ Network visibility module Access managers
- ☒ Secure Firewall posture
- ☐ Umbrella
- ☐ Ege posture

☒ Diagnostic & Reporting Tool

Install scheduled

ProgramData \ Cisco \ cisco Secure client \ Umbrella \ data

Step-2

DNS policies

identities >

Roaming

~~Desk~~ Computers >

Select the user

Lec-35: Mobile Agent overview



1

Root certificate of umbrella

iOS

Android

BYOD

Full management

Bring your device / MDM or Intune

Compass mentalization

1 Configuration

2 Policy

3 Assignment

4 Users

Set-1

Core identities

Deployment > Mobile devices > setting >

Failopen - umbrella unreachable, but allow users to do something in mobile

Fail close - Block everything when umbrella is unreachable

Administrator email - for resolution purpose

Device Identification - Full identifier

User Notification - Never

Download Android/mac config file.

Go Enterprise ^{admin} portal > Apps > +Add
Deploy an app

managed by Google play APP
> Cisco secure client ~~or~~ Anyconnect

St-2

Apps > App configuration policies > Add > mobile device >

Create APP configuration policy

Name: anything

Platform: Android Enterprise

Targeted app: Cisco Secure client

Next > Setting

Configuration settings: Use configuration designers

Configuration key:

value

value

☒ Umbrella Organization Id

☒ Umbrella Registration Token

Next > Assignments

Add groups

Umbrella client

St-3: Umbrella CA certificate is installed/trusted on mobile devices

Devices > Configuration profiles

Create profile ~ Android Enterprise

Trusted certificate

(Steps)

Lec-38: IOS Agent Deployment:

Deployments > Core Identities > IOS > Administrator Email
download

Str-2

Microsoft Intune

XML Profile

Devices > Configuration Profiles (Create profile)

Profile type: Template

(Template name: custom)

Basics

Name: anything/umbrella-F6

Configuration settings

Custom Configuration profile name: _____

Configuration profile file: XML

Assignments:

add group (umbrellaClient)

Create a profile → Template (Trusted Certificate)

Root CA Certificate of umbrella

Included groups

Add groups → umbrella client

Usage

Apps > IOS | Spadocore provisioning profiles > Select app type

IOS Store app

Also security connector

Lec-39: Mobile Agent Deployment Validation

MS policy >

ADV settings

Intelligence proxy

SSL navigation

> Mobile Devices >

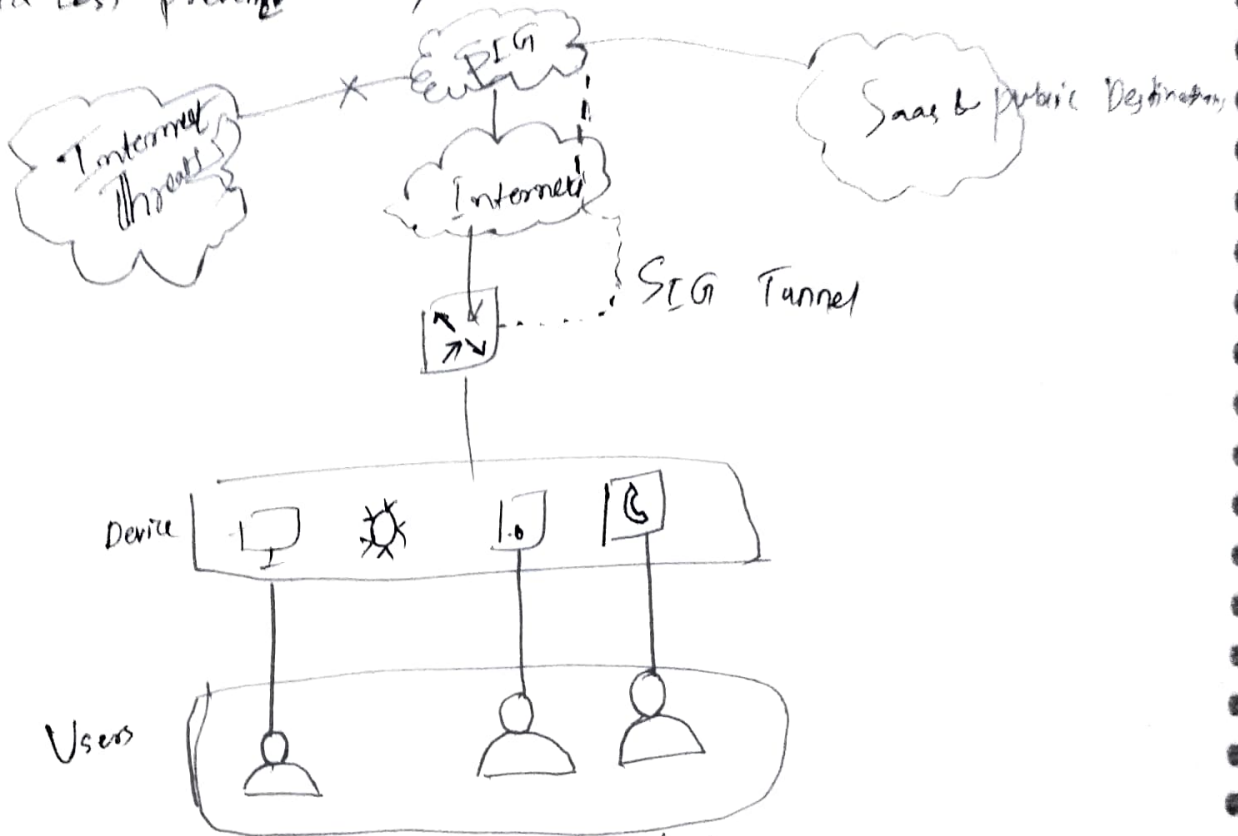
Security settings

High
Medium
Low
Custom

Application settings

Session 3 Lec 40: SIG Introduction

- i) Additional Inspection
- ii) cloud delivered Firewall
- iii) Intrusion prevention
- iv) Secure Web Gateway
- v) File & malware protection
- vi) cloud access security Brokers (CASB)
- vii) Data Loss prevention
- viii) 146.118.0.0/16 and 155.190.0.0/16



- SIG STEPS:
- (i) DNS layer security (iv)
 - (ii) cloud-delivered firewall (ips inspection)
 - (iii) Secure Web Gateway (url)
 - (iv) CASB - Secure Apps
 - (v) Interactive Threat intel - Security investigation

LEC 41: SIG use cases:

DNS Security (Essential or Advantage)

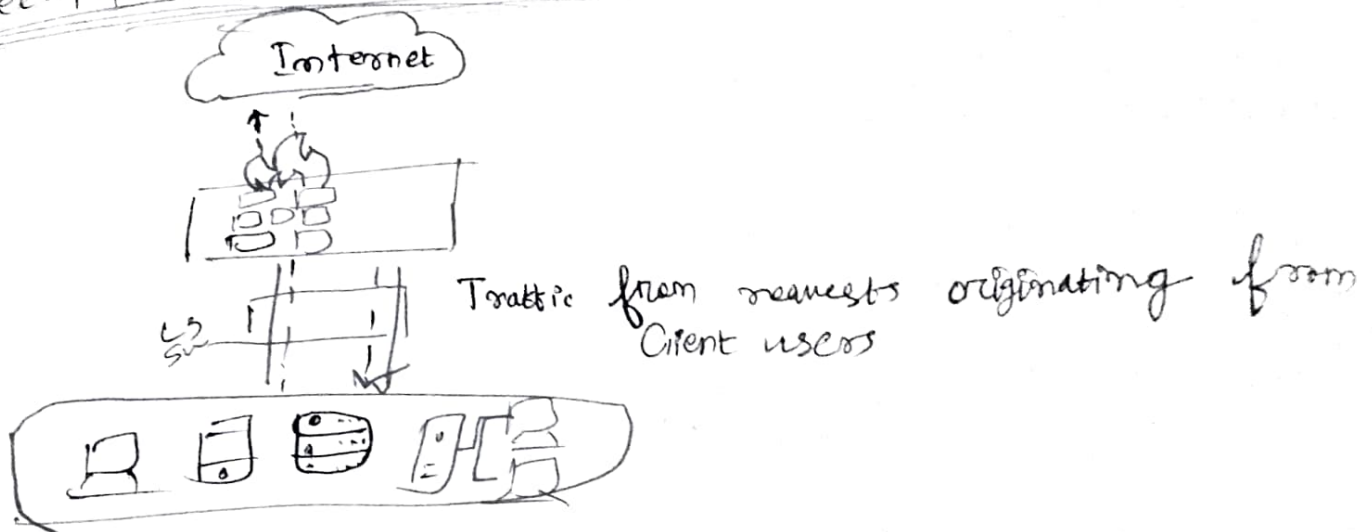
- Large Enterprise
- Global presence and 100's office
- Already have NGFWs, XDR, SIEM in place

Secure Internet Gateway (SIG)

- Small to medium Business
- Regional presence and smaller footprint
- No NGFWs, XDR, SIEM

Lec 3:11 - SEC - Cloud Delivered Firewall (CDF)

Lec-49: SEC - CDF Introduction



Umbrella Cloud Delivered Firewall

- (i) No physical Firewall at site
- (ii) Cost Effective Solution
- (iii) Requires Gateway that is compatible with SaaS VPN Tunnel
- (iv) NATs to 146.112.0.0/16 and 155.190.0.0/16
- (v) Layer 3 and Layer 7 Aware

Lec-43: SEC - CDF Tunnel Setup

Deployment > Core Identities > Network Tunnel > Add a New Tunnel

Tunnel Name: Secure-BranchOffice Tunnel

Device Type: (Remote Device)

Tunnel ID and passphrase

Site

Routing

Step 2

Site to mesh and create a tunnel

Security & SD-WAN > Site-to-site VPN > HUD

VPN Setting

Local Subnet

VPN
Variable

Anyconnect disch

NAT Traversal > 0 Automatic

Organization-wide Settings

Non-mesh VPN peers

Name

Securewise-office-tunnel

ike version

ipsec policy

2

Choose

Preset

Pre-shared key

Secret key

Public IP

Local IP

Generated Tunnel IP

Remote IP

Private Subnet

0.0.0.0/0

Availability

All networks tied to Securewise-office-network

Lec-44 - SIG CDF Policies

Policy > Management > Firewall policy

Add >

Rule name: Block SSH

Priority Order: 1

Rule action: Block

Rule criteria:

Rule Criteria:

Protocol: TCP, Sources: Securewise-office-tunnel

CIDR IP Address

Any

Ports: any, Destinations: CIDR IP Address: Any & Ports

Application

Rule schedule: , Hit Counter: , Logging:

Lec-45: SIG - CDF Intrusion prevention

How we can enable Intrusion prevention and applied on Cisco Umbrella (SIG)

9) Policy > Management > Firewall policy > Configure > Settings on

Intrusion System Mode

- Detection (IPS)
- Protection (✓) EPS

Apply to IPS Signature List

- Connectivity over security
- Balanced security and connectivity (✓)
- Security over connectivity
- Maximum Detection

S4-2 Policy components > IPS signature lists

For New - ADD > signature List Name

IPS signature List Template

Section-12 - Secure Web Gateway

Lec-46: SWG Introduction:

i) module of the SIG offering

ii) modes of deployment

• PAC file • proxy chaining

• IPsec tunnel • Anyconnect roaming

Security module

iii) Define acceptable use policy

iv) Define privacy policy

Modes of deployment

For Roaming users / Clients

not recommended

where is your proxy Server with Port no (more than one proxy WSA + Umbrella)

SWG is also known as Full proxy in umbrella.

URL logging and
Real-time
reporting

File inspection,
Sandboxing &
blocking

Secure web Gateway

Content &
app control

SSL traffic decryption
&
inspection

Lec-47: SWG Deployment

Creating web policies

- i) Web policy Applies to SWG components only
- ii) Web policy is made up of one or more Rulesets
- iii) Each Ruleset and it's Own Rules
- iv) Decryption of web traffic Requires Umbrella Root CA installed
- v) Rules are tagged with priority
- vi) Highest priority Rules are placed at the Top
- vii) Evaluation is in Descending Order and Stops if a match is found

Policy > management > web policy > Add

Ruleset name

Ruleset Identifiers

Block Page

Tenant control -

File Analysis - (Threat Grid malware analysis)

→ Sandboxing

Filetype Control - (Don't allow users to download exe files)
HTTPS inspection / SSL decryption - decrypt the https traffic
(needs umbra Root CA certificate)

PAC File - URL of proxy file

Safe Search -

Ruleset Logging -

SAML -

Security Setting -

ADD Rules -

<u>name</u>	<u>Action</u>	<u>Identify</u>	<u>Category</u>
Blockwise-Rule-1	B/A		

Rule configuration

Time (Any day,)

Lec-48: SWG Testing:

Practice
Download a file,

Gaming Rule block
↓

Rockstar games,

Lec-49: SWG Data Loss prevention:

It is a process used to identify certain part of critical info like credit card no, ^{we want to make sure the} any private info exchanged then we need to know.

ex- Credit card on spread sheet exchanged from org to internet

Policy > management > Data Loss prevention > ADD Rule

i) Add a real time Rule (name, severity)

ii) Data classification  name, Description (opt)

Select Boolean operators
 OR • AND

Select data identifiers

- ☐ Social Security Number (ID) - Strict
- ☐ Credit Card Number (IMP) - Strict

Custom data identifiers

Regex" . . .

iii) Add New Real Time Rule

Rule Name - Securewire-DLP

Severity - High / med / low

Data Classification

- Content
- File Name
- Content File Name

(iv) File Labels (includes critical documents - uses file header)
 (opt)

(v) Identifiers - (identity - tunnel / Network / user)

Destinations

• All Destinations

• Select Destination Lists and Application for ~~function~~ inclusion

□ Select destination List and Application for Exclusion
(opt)

Action
└─ Monitor
└─ Block

Lec-50: SIG - SWG DLP SaaS API Setup

DLP - private information

what~~is~~ about if you already have things like Microsoft 365, OneDrive, Google Drive, and if you want to scan those environments to find out if your employees or users have any sensitive information stored in their unencrypted

Configuration :

Policies > Management > Data Loss prevention policy >

Add Rule > SaaS API Rule > Go to authentication page >

* Platform (Dropbox, Box, Microsoft 365, Webex Teams, Google)

Suppose

Microsoft 365

Cloud malware

DLP - Authorize New Tenant

□ Global admin

□ Sharepoint online

□ Firewall rules prevent third party - applications

below steps

(i)

Name (ii)

Name ~~the~~ Tenant

Tenant name

(iii) Microsoft Credentials

ask permission requested

* File owners

OR
OSPER

use

* Exposure

Select which file sharing exposure to consider in addition to this rule's other settings

☒ Sharing publicly ☒ Shared with external users ☒ Domain-wide share

☒ Shared with internal users ☐ Shared with specific users

Action:
Quarantine (✓)
Revoke
Monitor

(Study
False Positive
False Negative

Discovery Scan :

Lec 51 - SIG - SWG SAML Setup

(Before accessing anything, we need to provide username & password)

Cisco Umbrella is giving you secure web gateway as a proxy to access the internet resources in different URLs,

If you wanted to authenticate users before they even get to the internet.

that feature available through (SAML)

Policy > management > web policy > rule set setting

enable SAML

(i) SAML Metadata

(create Access key & token)

Set up umbrella as an enterprise application (ii)

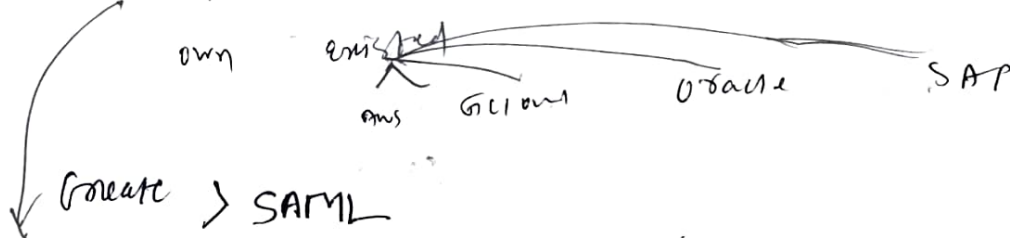
on our identity provider, which happens to

be Microsoft 365 Azure AD (Other idp)

Azure ID

Enterprise Applications

> New application >



Add identifier

(get in umbrella)

copy URL

(umbrella ID)

SAML certification

Federation metadata: (download XML)

get it from

Users and group: add user

Umbrella

Deployments > Configuration > SAML configuration

(i) Add > Select an identity provider

• AD FS • Azure AD Security • OpenAM • PingID
• Other

Organization-specific Entity ID:

☒ Organization Disabled

(ii) Configure Azure metadata

• XML, file upload

• Manual configuration

(iii) Download the Umbrella metadata file

(iv) Configure

upload the Azure AD XML file

(v) Re-authentication proxy user

Sec-13: Umbrella Investigate

Lec-52: Using Investigate

Investigate < Search
pattern Search

It is a feature inside Cisco Umbrella that helps security teams analyze and investigate suspicious domains, IP addresses, and URLs.

Lec-53: Investigate API

Investigate > API keys > Create New Token

It gives you visibility into:

- Domain details
- IP address information
- DNS Request history
- Security risk level
- Threat intelligence data

go to

Cloud Security API

```
import investigate  
api_key =
```

```
inv = investigate.Investigate(api_key)
```

```
inp = input("Enter the main name: ")
```

```
Print ("### Categorization and Status ###\n")
```

```
Print (inv.categorization(inp))
```

```
Print ("\n\n### Risk Score ###\n")
```

```
Print (inv.risk-score(inp))
```

```
Print ("\n\n### Score result ###\n")
```

```
Print (inv.security('test.com'))
```

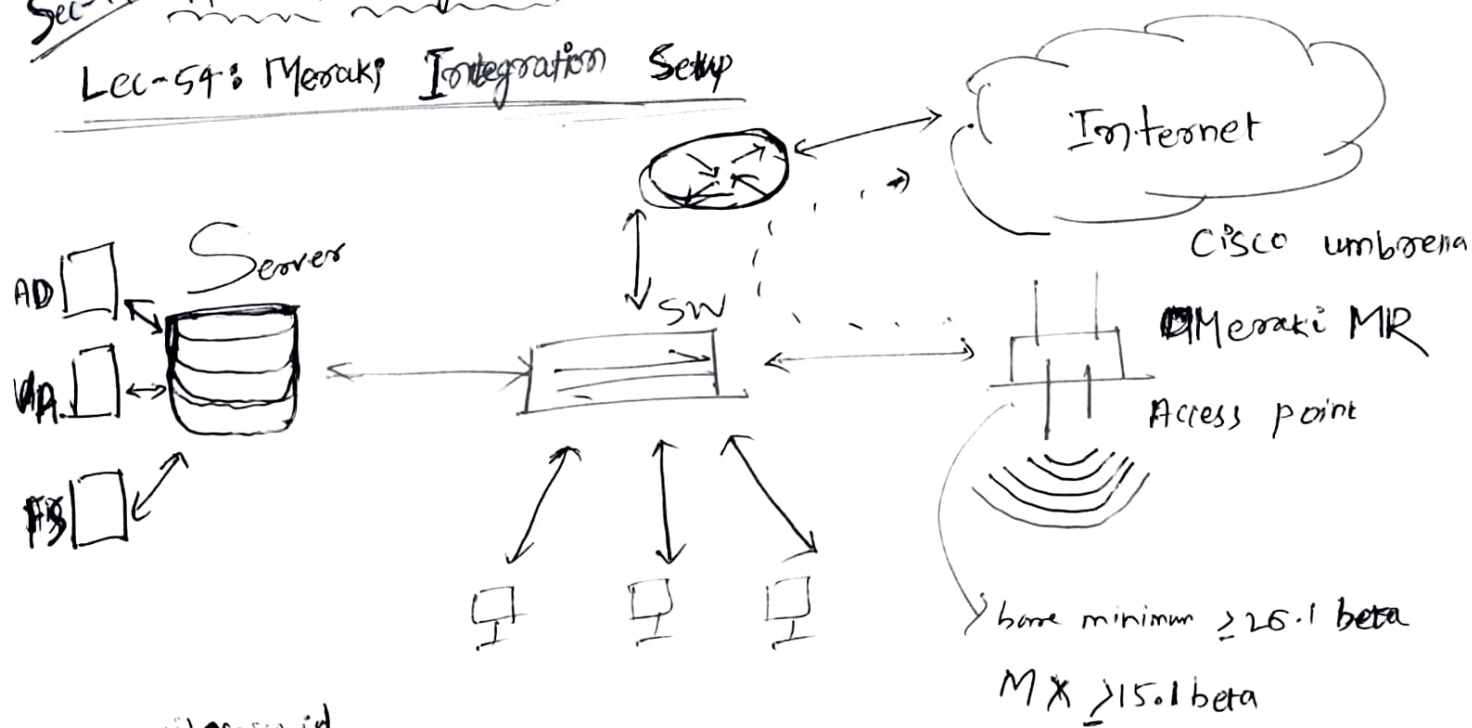
```
Print ("\n\n### Domain Tags ###\n")
```

```
print (inv.domain-whois(inp))
```

(we can use function that referent to us)

Sec-14: Advanced Integrations

Lec-54: Meraki Integration Setup



- (i) Office id
- (ii) Guest id

(i) client connect to id that we have the access point would add an identifier in those DNS packets that are received from these clients for resolution and what it would do is it encrypt these packets and send it out to cisco cloud umbrella cloud. (DNS script)

(ii) Based on identifiers, it checks the policy and take action like permit or deny, to send out the IP address that is resolved from the domain, or it would send out the IP address or the block page if it is blocked.

(iii) So once the response comes back, it would be sent out to the Meraki ^{MR AP} ~~AP~~, which would subsequently send out to the client.

(iv) Now when the communication is moving out from Meraki, the DNS request would be sent out to the management IP or the MR AP and they send out to the cisco umbrella cloud.

(Note: Here Virtual Appliance don't play any role out here)

✓ to keep it secure

Wireless > Config > Firewall & Traffic Shaping > Block app & Content Categories

SSID ← Office
Guest

DNS Layer (default) Protection
(Choose relevant policy)

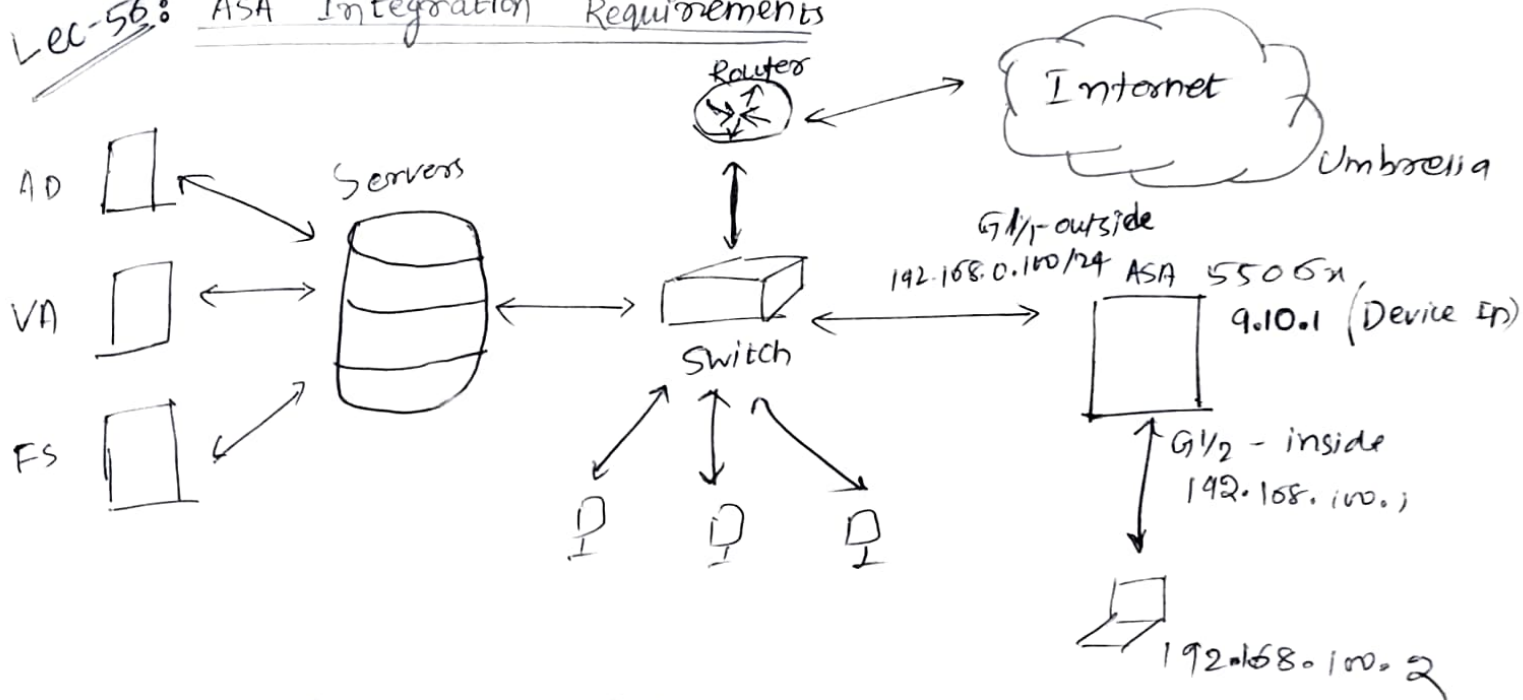
☒ Enable Umbrella Protection

After that check in the deployment ^{umbrella secret} > Core identities & Network devices
(homework:)

Lec-55 : Measari Integration testing

Content Setting Appliat - MR - Guest
Application

Lec-56: ASA Integration Requirements



1. SMB, MB ... cost restrictions on servers

2. Old generation of ASAs

ASA adds device id when ASA device to umbrella

packets pass through from

Lec-57: ASA Integration & Configuration

ASDM

(co) # dns domain-lookup outside

(co) # dns name-server 208.67.222.222 208.07.222.220

(co) # crypto ca trustpoint cxi

(co-ca-trustpoint) # enrollment terminal

crypto ca authenticate cxi

Connectors - It connects ASA to cisco umbrella

(co)

umb

Admin > API keys (Created Key Similarly ^{that} created for cisco umbrella)

Select > Legacy Network Device > Create

(co) # umbrella-global
token (token number got from umbrella)
ASA
(co-umbrella) # public-key (public key)
timeout eds 00:01:00
exit

enable umbrella in the
DNS inspection policy > enable this globally by adding it to the
DNS map of the traffic inspection or
the policy map we have

ASA

(co) # policy-map type inspect dns present-dns-map
(co-map) # parameters
(co-map-p) #

Umb

Policies > Management > All policies

ASA

umbrella tag default-policy
dnscrypt

Umb

Deployments > Core Identities > Network Devices

we can see ASA device here

All policies

edit identities

o Network Devices

ID Office AP

ID Scanwire - ASA01

format
(c) # exit

(c) # show ^{view} service-policy inspect dns

Umbrella resolver mode in ASA

Fail-open

Fail-close

(c) # policy-map ^{type} inspect dns ^{Changes} pset-dns-map

(c-pmap) # parameters

(c-pmap-p) # umbrella ^{exit} fail-open tag SecureWire-Network-policy

(c) # dns name-server 8.8.8.8
 dns name-server 4.2.2.2

~~Umbrella~~ IF ASA isn't able to reach out to cisco umbrella
for some reason then the ASA will fall back to
8.8.8.8 or 4.2.2.2 DNS servers.

Lec-58: ASA Integration Validation

Deployment > Core ID > Network Devices

Lec-59: SIEM Integration

Admin > Log Management

- Amazon S3 -
 - use your company-managed Amazon S3 bucket
 - use Cisco-managed Amazon S3 storage

Select ~~an~~ region
US West (N, California)

Select a Retention Duration
30 days

[Secret Key]

SS browsers.com - test & validate S3 bucket

Download

Add New Account

Account Name: umbrella

Account Type: Amazon S3 Storage

Access Key ID: } can get it from umbrella dashboard

Select access key: }

Add External Bucket > Bucket name > [dnstlogs
proxylogs] Folders

download any file

.csr.gz ← 12/05/2017

→ proxy logs of the devices

Lec-60 - Security Event Reporting Integration

API Keys

It can get blocked requests and feed it to the sim. ~~Please~~ This method only gives the requests that are blocked, whereas the other method shows both blocked and permitted requests.

Create > Umbrella Reporting > gives
Key:
Secret:

Umbrella.logs.py :

import requests

import json

from datetime import datetime

import base64

APIkey = "..."

APIsecret = "..."

APIcombined = APIkey + ":" + APIsecret

base64_encoded = base64.standard_b64encode(bytes(APIcombined, 'utf-8')).decode('utf-8')

organizationID = "..."
reporting-url = "..." + organizationID + "..."

```
headers = {  
    "..." : "..." + base64-code  
}
```

```
result = requests.get (reporting-url, headers = headers)  
outputFinal = result.json()  
customTime = str(datetime.utcnow())  
customTime = customTime.replace(':', '-')  
myFile = open ("UmbrellaLogs" + (str(customTime)) + '.txt', "a")  
Print (outputFinal)  
myFile.write (str(outputFinal))  
myFile.close()
```

Sec 15: Administration

Lec 61: Portal Administration

Admin > Accounts >

Create account with some level of privilege

- Full admin - Access to everything, including the ability to create new roles, view all the pages and reports in the dashboard, bypass users, not - policy settings
- Read only - Access to only view
- Block page Bypass -
- Reporting only = Access to view and run reports

Add - new role (individually provide access to different components)

Logmanagement - Change location (Storing ^{log} data in a different location)

Lec-02: Basic Troubleshooting

Overview Tab

If you can't see the device in Roaming Computers, It could be most probably be an issue with the agent by itself, whether it's powered down or the machine itself is offline or if there is any (FW) or any other content blocking filter that is blocking certain ports.

Roaming computers

□ Win7

⊙ Offline

DNS Layer Encryption disabled

12 hours ago

It is a VM, we need to contact owner of the machine to get it online.

Mobile Devices or MDM

Esxi > vm > umbrella VAA (If you see Local DNS Fails)

Ping the VAI - if ping unreachable

~~Cisco PACE~~

if problem with other factors than the VA by itself ^{is} not able to get to the ^(update, umbrella cloud) internet, then we need to check

from VA to the internet connection.

(*) The second issue that normally shows up and I have seen this very common is when you add an IP address for your public interface out here, it doesn't come online, so it doesn't even go from inactive to active.

Most of the time when this happens is if you have multiple subscriptions. So if you were a customer that was using open DNS and you used this IP address as an egress point out there on the open DNS console, what would happen is it wouldn't allow you to have duplicate entries. So the new network that you put up here would never come up until and unless you delete the old entry which is present on any other open DNS or umbrella subscription, which is other than this current organization. So that is the one of the common issues that I've faced, is when users sign up for trial or company sign up for trial, they sign up this account or (this network here) and you forget to delete this

when the trial is over. So when you spin up a new instance, bringing this IP back online on the new console becomes very difficult.

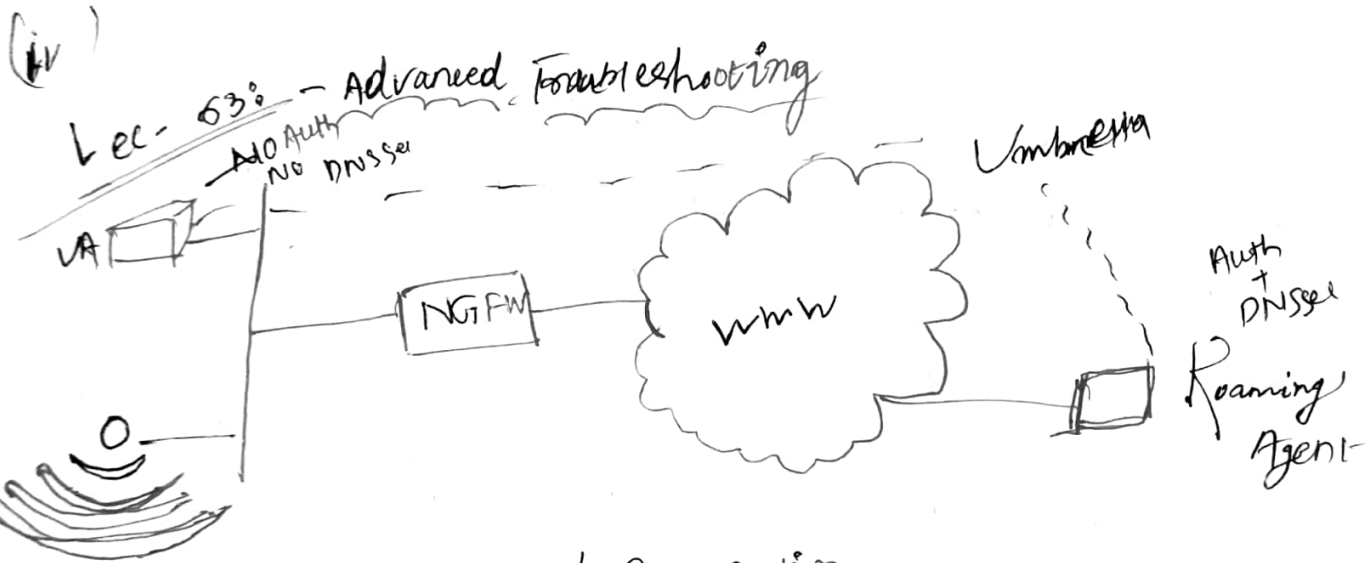
So in that event, we need to contact the Cisco Tech and they would help you delete it from their backend and you can import it in the new one.

Sometimes what happens is if you do a bulk import here. If you set up multiple IP addresses and for some reason you aren't coming out from that IP address. ~~Now in the~~ Let suppose

here I'm coming out from 13.14.1.164.158 IP, which is the same as that is set up here. However, if I was coming out from, let's say 1.2.3.4, and if I try to enter this IP address, it wouldn't allow me to do so, but Cisco orient to protect.

Cisco wants to authorize specific owners of the IP ranges so that they know that you're not entering somebody else's IP address here for DNS inspection. (Cisco Tech)

(iii) Certificate is n't trusted - root CA Certificate must be installed on agent devices



Sec-16: Reports and Investigation

lec- 64:

Core Reports > Security Overview

Core Reports > Additional Reports